AWS CloudFormation — Important Interview Questions & Answers

1.What is AWS CloudFormation?
AWS CloudFormation is an AWS service used to create and manage AWS resources using templates.
Instead of creating resources manually, we can define them in a template and CloudFormation creates them automatically.
Simple ga gurthupettuko:
CloudFormation ante template use chesi AWS resources ni automatic ga create and manage cheyyadam.
Eng: CloudFormation is used to create and manage AWS resources using templates.

2.What is a CloudFormation Template?
A CloudFormation Template is a YAML or JSON file that defines the AWS resources and their configuration.
It describes what resources should be created and how they should be configured.
Simple ga gurthupettuko:
Template ante AWS resources ni define chese YAML or JSON file.
Eng: A CloudFormation Template defines AWS resources and their configuration.

3.What is a CloudFormation Stack?
A Stack is a collection of AWS resources created and managed together by CloudFormation.
For example, one stack can contain an EC2 instance, security group, and other required resources.
Simple ga gurthupettuko:
Stack ante CloudFormation tho create and manage chese AWS resources collection.
Eng: A Stack is a collection of AWS resources managed together by CloudFormation.

4.What are Resources in CloudFormation?
Resources are the AWS services or components that CloudFormation creates and manages.
Examples include EC2 instances, S3 buckets, VPCs, IAM roles, and security groups.
Simple ga gurthupettuko:
Resources ante CloudFormation create chese actual AWS services or components.
Eng: Resources are AWS components created and managed by CloudFormation.

5.What are Parameters in CloudFormation?
Parameters allow us to pass values into a CloudFormation template when creating or updating a stack.
They help reuse the same template with different values such as instance type, VPC ID, or environment name.
Simple ga gurthupettuko:
Parameters ante template ki outside nundi values pass cheyyadaniki use chestham.
Eng: Parameters allow us to pass input values into a CloudFormation template.

6.What are Outputs in CloudFormation?
Outputs allow CloudFormation to display useful information after a stack is created or updated.
For example, we can output an EC2 instance ID, load balancer DNS name, or resource ARN.
Simple ga gurthupettuko:
Outputs ante stack create ayyaka useful resource information display cheyyadaniki use chestham.
Eng: Outputs display useful information from a CloudFormation stack.

7.What is a CloudFormation Change Set?
A Change Set shows the changes that CloudFormation plans to make to an existing stack before applying them.
It helps us review the impact of a template change before updating the stack.
Simple ga gurthupettuko:
Change Set ante stack update mundu em changes jaruguthayo preview chuse option.
Eng: A Change Set previews the changes before updating a CloudFormation stack.

8.What is CloudFormation Drift Detection?
Drift Detection checks whether the actual configuration of resources has changed from the configuration defined in the CloudFormation template.
It helps identify manual changes made outside CloudFormation.
Simple ga gurthupettuko:
Drift Detection ante template lo unna configuration ki actual resource configuration same ga unda leda check cheyyadam.
Eng: Drift Detection identifies differences between the template configuration and actual resource configuration.

9.What happens when we update a CloudFormation Stack?
When we update a stack, CloudFormation compares the existing stack with the new template or parameter values.
It then creates, updates, or replaces resources as required by the changes.
Simple ga gurthupettuko:
Stack update chesthe CloudFormation changes ni identify chesi required resources ni update or replace chestundi.
Eng: CloudFormation updates or replaces resources based on the stack changes.

10.What happens if CloudFormation stack creation fails?
If stack creation fails, CloudFormation reports the failure and can roll back resources depending on the stack configuration.
This helps avoid leaving the environment in an unwanted partial state.
Simple ga gurthupettuko:
Stack creation fail ayite CloudFormation rollback chesi created resources ni remove cheyyachu.
Eng: CloudFormation can roll back resources when stack creation fails.

11.What is CloudFormation Rollback?
Rollback means returning the stack to its previous stable state when an operation fails.
It helps prevent incomplete or inconsistent infrastructure after a failed deployment.
Simple ga gurthupettuko:
Rollback ante failure vachinappudu stack ni previous stable state ki teesukelladam.
Eng: Rollback returns the stack to a previous stable state after a failure.

12.What is Infrastructure as Code?
Infrastructure as Code means defining and managing infrastructure using code or configuration files instead of creating resources manually.
CloudFormation is an AWS service that supports Infrastructure as Code.
Simple ga gurthupettuko:
Infrastructure as Code ante infrastructure ni manual ga kakunda code/template dwara manage cheyyadam.
Eng: Infrastructure as Code manages infrastructure using code or configuration files.

13.What are CloudFormation Nested Stacks?
Nested Stacks allow one CloudFormation stack to create or manage another CloudFormation stack.
They are useful for organizing large templates into smaller reusable components.
Simple ga gurthupettuko:
Nested Stack ante oka CloudFormation stack lopala another stack use cheyyadam.
Eng: Nested Stacks allow CloudFormation stacks to be organized into smaller components.

14.What is the difference between CloudFormation and Terraform?
CloudFormation is an AWS-native Infrastructure as Code service mainly designed for AWS resources.
Terraform is a multi-cloud Infrastructure as Code tool that can manage resources across AWS and other cloud providers.
Simple ga gurthupettuko:
CloudFormation mainly AWS kosam, Terraform multiple cloud providers kosam use cheyyachu.
Eng: CloudFormation is AWS-focused, while Terraform supports multiple cloud providers.

15.How would you create an AWS infrastructure using CloudFormation?
I would first create a YAML or JSON template defining the required resources and their configuration.
Then I would create a CloudFormation stack using the template and provide any required parameters.
Simple ga gurthupettuko:
Template create chesi → Stack create chesi → Parameters provide chesi infrastructure deploy chestham.
Eng: I would define resources in a template and create a CloudFormation stack from it.

16.How would you troubleshoot a CloudFormation stack failure?
I would check the stack events to identify which resource failed and the reason for the failure.
Then I would check the template, parameters, IAM permissions, dependencies, and resource configuration.
Simple ga gurthupettuko:
Stack fail ayite Events → Failed Resource → Template → Parameters → IAM Permissions check chestham.
Eng: I would check stack events, the failed resource, template, parameters, and permissions.

17.What is the most important thing to remember about CloudFormation?
CloudFormation is an Infrastructure as Code service used to create and manage AWS resources through templates.
The important concepts are Templates, Stacks, Resources, Parameters, Outputs, Change Sets, and Drift Detection.
Simple ga gurthupettuko:
CloudFormation ante template use chesi AWS infrastructure ni automatic ga create and manage cheyyadam.
Eng: CloudFormation is used to create and manage AWS infrastructure using templates.